KUANG-CHENG, CHENG

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EDUCATION

Sep. 2021- May. 2023 **University of Wisconsin-Madison, Madison**
M.S. in Geographic Information Systems and Cartography
- GPA: 3.83

Sep. 2011- Jun. 2015 **National Central University, Taoyuan, Taiwan**
B.S. in Civil Engineering
- EAC accredited program.
- GPA: 3.34.
- Relevant coursework:
  (1) Programming for Engineering (C++) using Visual Studio.
  (2) Numerical Analysis: MATLAB Program design.
  (3) Engineering Statistics.

WORK EXPERIENCE

Apr. 2024- Jul. 2026 **GIS Analyst**
Houston Public Works *Houston, TX*
- Developed spatial data pipelines with ModelBuilder and refined exported Python scripts to enhance automation, reduce errors, and improve processing efficiency.
- Created, maintained, and updated geospatial data in the ArcGIS Enterprise database.
- Performed data QA/QC and ETL (extract, transform, load) on Excel-based GIS datasets, standardizing workflow to support enterprise data governance and improving data quality.
- Created maps to support various engineering projects.
- Coordinated and managed the GIS projects for engineering and IT groups to ensure data integrity and determine project priorities.
- Translated ambiguous business requests into structured spatial deliverables, designing geodatabase templates and datasets for internal use and web publication.
- Designed and developed a dynamic Excel dashboard to automate data analysis and visualize key metrics for reporting and operational discussions
- Analyzed different versions of FEMA floodplain data and produced comparison maps using distinct symbology to display changes in floodplain boundaries.

Sep. 2023- Apr. 2024 **GIS Volunteer**
URISA's GISCorps *Des Plaines, IL*
- Test and help with the development of the user interface (UI) of Underpass, a ReactJS library, for the Humanitarian OpenStreetMap Team.
- Collect street-level images in support of the creation of smart map data for Mapilio, a map platform.
- Edit buildings on satellite photos for Humanitarian OpenStreetMap Team projects.

Jan. 2022- Aug. 2022 **GIS Metadata Technician**
University of Wisconsin-Madison *Madison, WI*
- Assessed LiDAR data inventory statuses.
- Created original ISO metadata for nearly 400 new Wisconsin county datasets.
- Added interactive tile indexes for Wisconsin LiDAR-derivative data resources.
- Edited Wisconsin V8 county parcel data.

Aug. 2019- Jul. 2021 **Assistant Engineer**
Ministry of Transportation and Communication *Taipei, Taiwan*
- Coordinated the COVID-19 mask distribution project to ensure companies and agencies had sufficient supplies to maintain workplace safety.
- Processed COVID-19 mask demand data and monitored logistics to ensure timely and accurate distribution.
- Updated, maintained, and delivered routine mobilization data.

- Evaluated mobilization task guidelines for air, shipping, and road transportation.
- Arranged the annual training program for senior officers and project leaders.

Nov. 2016- **Junior Engineer**
Jul. 2021 *Freeway Bureau* *New Taipei, Taiwan*

- Review and revise environmental impact assessment reports, ensuring they are aligned with policies and regulations.
- Digitized Taiwan's freeway historical documents for historical data preservation.
- Held environmental education events, evoking awareness of environmental issues.
- Examined design drafts and contracts of national bridges.

Jul. 2014- **Engineer Intern**
Aug. 2014 *CECI Engineering Consultants, Inc.* *Taipei, Taiwan*

Gained experience at the most prominent civil engineering design company in Taiwan.

- Visualized civil engineering project drafts and data on Google Earth.
- Completed sectional drawings of roads on the mountain area of Tai-20 (one of Taiwan's provincial highways) using AutoCAD.

SKILLS

Languages: Mandarin Chinese (native), English (fluent).
Computer Software: AutoCAD, SketchUp, ArcGIS Pro, QGIS, MS Power BI, Adobe Illustrator, MS Office (including Word, Excel, PowerPoint)
Programming: Python, SQL, HTML/CSS/ JavsScript, GitHub
Cloud Computing Environment: Google Cloud Platforms, ArcGIS Online (AGOL)
Database System: PostgreSQL (with PostGIS), MongoDB, Snowflake
Other: Multi-tasking, Time Management, Problem-solving, Project Coordination, Supervisory Skills, Data Management, Communication

GIS PROJECTS

- **Spatial Big Data Analysis & Visualization:**
  "Madison Metro Transit Redesign Evaluation" - Perform data ETL, statistical data analysis and visualize spatial data on an interactive map using Python in Jupyter notebook.
- **Spatial Database Design and Management:**
  "US National Parks Recreational Database" - Create database to integrate visitation records, facilities, and spatial data inside or nearby national parks using PostgreSQL (with PostGIS extension).
- **GIS Analysis:**
  "Road salt use impact on major sensitive areas in the metropolitan area of Marathon County"-Perform spatial analysis and map the potential contamination strength on each road for marathon county transportation planner.
- **Geovisualization and UI/UX Design:**
  "United States Exports By Industry" - Create a web-based user interface to visualize the data collected from the Observatory of Economic Complexity (OEC) using JavaScript/HTML/CSS with Leaflet library.
- **GIS Development Practicum:**
  "Yearly ridership data analysis and visualization" - create a workflow for data ETL, open data publishing, and spatial data management and visualizations.
- **Cartography (Mapping) Design:**
  "Taiwan Public Transportation Infrastructure – bus ridership vs. population"-Aggregate data from multiple open data resources and visualize data on a map using ArcGIS Pro and Adobe Illustrator.

CERTIFICATION

2016 **Professional and Technical Personnel Examinations of Civil Engineers, Taiwan**

- Certificate issued by Ministry of Examination, R.O.C. (Taiwan).
- Officially qualified to be a Professional Civil Engineer (PE) in Taiwan.
- Qualification Rate (2016): 11.53%.

2016 **Civil Service Level Three Senior Examination in Civil Engineering, Taiwan**

- Certificate issued by Ministry of Examination, R.O.C. (Taiwan).
- Passed the public servant exam as a Junior Engineer.
- Qualification Rate (2016): 25.78%.